

## Course Outline

### 1. Course Identity

#### A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course code             | MDS5115                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Course title (English)  | Big Data Modeling and Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Course title (Chinese)  | 大数据建模与管理                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Units                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Language of Instruction | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Description (English)   | As the web technology and mobile use rapidly evolves, the volume of user-generated data expand exponentially. The distillation of knowledge from such a large amount of unstructured, dynamically changed data is an extremely difficult task without the help of distributed techniques. This course introduces most state-of-the-art Big data analytical concepts, techniques and tools. By taking this course, students will gain hands-on technical experiences on solving Big Data problems using distributed algorithms and tools widely adopted in industry. The topics include basic concepts about Big Data, installation and configuration of Hadoop and Spark under a multi-node environment, distributed algorithms (recommender systems, clustering, classification, topic models, and network analysis), web crawling and web data extraction using major application programming interfaces. |
| Description (Chinese)   | 随着网络技术和移动使用的迅速发展，用户生成的数据量呈指数级增长。如果没有分布式技术的帮助，从如此大量的非结构化、动态变化的数据中提取知识是一项极其困难的任務。本课程介绍最先进的大数据分析概念、技术和工具。通过本课程，学生将获得使用行业广泛采用的分布式算法和工具解决大数据问题的实践技术经验。主题包括大数据的基本概念、多节点环境下Hadoop和Spark的安装和配置、分布式算法（推荐系统、集群、分类、主题模型和网络分析）、网络爬虫和使用主要应用程序编程的网络数据提取接口                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

#### B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course in CUHK, please make multiple copies of the block below.

|                        |  |
|------------------------|--|
| Course code            |  |
| Course title (English) |  |
| Course title (Chinese) |  |
| Units                  |  |
| Description (English)  |  |
| Description (Chinese)  |  |

## 2. **Prerequisites / Co-requisites**

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen\* or any other requirements (e.g., having taken certain subjects in high school).

(\* Because course codes may not yet be stable, please provide both course code and course title.)

### A. **Prerequisites**

### B. **Co-requisites**

## 3. **Learning Outcomes**

After learning this course, students can master the basic techniques and tools to handle the large volume of data arising from the practice or the research problem.

## 4. **Course syllabus**

the same as course outline

## 5. **Assessment Scheme**

| Component/ method    | % weight |
|----------------------|----------|
| midterm presentation | 40       |
| final project        | 55       |
| class engagement     | 5        |

## 6. Grade descriptor

### General

| Grade | Description                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | Outstanding performance on all learning outcomes.                                                                                                                                                            |
| A-    | Generally outstanding performance on all (or almost all) learning outcomes.                                                                                                                                  |
| B     | Substantial performance on all learning outcomes, OR high performance on some learning outcomes which compensates for less satisfactory performance on others, resulting in overall substantial performance. |
| B+    | Substantial performance on all learning outcomes, OR high performance on some learning outcomes which compensates for less satisfactory performance on others, resulting in overall substantial performance. |
| B-    | Substantial performance on all learning outcomes, OR high performance on some learning outcomes which compensates for less satisfactory performance on others, resulting in overall substantial performance. |
| C     | Satisfactory performance on the majority of learning outcomes, possibly with a few weaknesses.                                                                                                               |
| C+    | Satisfactory performance on the majority of learning outcomes, possibly with a few weaknesses.                                                                                                               |
| C-    | Satisfactory performance on the majority of learning outcomes, possibly with a few weaknesses.                                                                                                               |
| D     | Barely satisfactory performance on a number of learning outcomes                                                                                                                                             |
| F     | Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.                                                                                           |

## 7. Feedback for evaluation

CTE

## 8. Reading

**Recommended Reading:**

## 9. Course components

| Activity                                     | Hours/week |
|----------------------------------------------|------------|
| Lecture (1.5 hour each, 2 lectures per week) | 3          |

## 10. Indicative teaching plan

| Week | Content/ topic/ activity                    |
|------|---------------------------------------------|
| 1    | Introduction to Big Data Analytics          |
| 2    | Memory Hierarchy                            |
| 3    | Introduction to Spark                       |
| 4    | Setting up Python with Spark                |
| 5    | Spark DataFrame Basics                      |
| 6    | Introduction to Machine Learning with MLlib |
| 7    | Linear Regression, Logistic Regression      |
| 8    | Decision Trees, Random Forests              |
| 9    | Classification                              |
| 10   | Filtering for Recommender System            |
| 11   | Natural Language Processing                 |
| 12   | Graph Neural Network                        |
| 13   | Final Project Presentation                  |
| 14   | Final Project Presentation                  |

## 11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

[Example: 150 students for lecture (x 2); 30 students for tutorials (x 10)]

## 12. Any other information

## 13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Have the details (as in this document) been approved at School or other level in CUHK-Shenzhen?

Submit application

**14. Version date**

|                |            |
|----------------|------------|
| Version number | 1          |
| As of (date)   | 2025-07-23 |